

Ablation across 6 datasets × 4 rts (seed=42). Top: rare F1. Bottom: FFR (max across rts).

V0 · baseline scANVI V2 · no necessity guard V4 · rank=2 fixed **V6 · full (ours)**
 V1 · no sep gate (LOW_SEP=0) V3 · rank=1 fixed V5 · no conformal τ V7 · rank=3 fixed (sensitivity)

